



Numerical simulation of molten pool flow behavior and keyhole evolution behavior in dual-laser beam oscillating bilateral synchronous welding of T-joints



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Highlights

- Dual-laser beam oscillating bilateral synchronous welding (DLBOW) is proposed, which has significantly reducing the porosity in T-joint welds.

- A numerical model of computational fluid dynamics has been developed which is in good agreement with the experimental results.
- The molten pool flow behavior and keyhole evolution under molten pool coupling were analyzed.

Abstract

In the context of global warming and energy shortage, the key means to increase product competitiveness is how to achieve lightweight manufacturing of large spacecraft skin-stringer structures by welding instead of riveting. In this work, we propose dual-laser beam oscillating bilateral synchronous welding (DLBOW) by combining dual-laser beam synchronous welding (DLBW) with the oscillating process to overcome the long-standing porosity problem in T-joint welds and investigate the effect of different laser incidence angles on the porosity of the weld during welding. At the same time, the corresponding numerical model was established by computational fluid dynamics to analyze the molten pool flow behavior and keyhole evolution under molten pool coupling. This is the first time that complete wire melting process has been achieved in the T-joint dual-laser beam synchronous welding numerical model, reproducing more realistically the influence of the wire fusion behavior on the molten pool flow. Both experimental and simulation results show that the porosity is significantly reduced under DLBOW compared to DLBW; in addition, the porosity is lower when the laser incidence angle is smaller. An in-depth analysis of the numerical model reveals the mechanism of this physical phenomenon: First, the oscillation process optimizes the laser energy density distribution and reduces the peak temperature of the T-joint molten pool. The impact of the metal flow on the keyhole is weakened, which reduces the probability of bubble formation and thus significantly decreases the porosity. Moreover, for the oscillation process, the keyhole effect is more obvious at large laser incidence angles, and the actual heat input to the molten pool is larger, which makes the slender keyhole more prone to collapse and more likely to generate pores. This work provides new insight into the interrelationship between keyhole and pore in T-joint dual-laser beam synchronous welding.

Keywords

Laser oscillating welding; Numerical simulation; Molten pool dynamics; Keyhole dynamics; Porosity

1. Introduction

Aluminum alloy is widely used in the aerospace industry due to its low specific gravity, high specific strength, and mature manufacturing and testing process[1], [2], [3]. The traditional skin-stringer reinforcement structure of the spacecraft wall panel is connected by riveting technology, which has shortcomings of large overall weight, limited processing efficiency, and higher manufacturing cost[4]. In the background of global energy shortage and environmental protection, the use of welding instead of riveting to achieve lightweight manufacturing of spacecraft wall plate structure is a critical tactic to promote product competitiveness[5]. Laser welding has high energy density, low heat input and excellent accessibility[6], [7], [8]. In the early 1990s, the German company Airbus pioneered the study of double-sided laser welding technology for the skin-stringer structures of aircraft fuselage panels, and applied the technology in the Airbus A318 and A380 fuselage panels [9]. Up to 20% weight reduction and over 41% manufacturing time saving without changing the structural design of the components [10].

However, for the special T-joint laser welding, the obstruction of the stringer makes the inclined state of the molten pool flow behavior and keyhole characteristics more complex, the welding process is prone to instability, bubbles escape path is restricted, resulting in the formation of a large number of macroscopic and microscopic pores in the welded joint [5,11]. The presence of pores may reduce the strength and fatigue properties of welded joints [12,13], which is not conducive to the widespread use of laser welding technology for T-joints.

The pores were classified by synchrotron X-ray imaging observations according to the location where they were generated[14]: (i) instantaneous bubbles formed in the post-keyhole, which tend to disappear rapidly after generation (ii) pore formed at the end of the laser path, which are removed by post-process polishing (iii) pore formed due to keyhole fluctuations at the bottom of the molten pool. Since the first two types of pore are less frequent and have less impact on the performance of the welded joint, the focus of research tends to be on the third type of pore. Generally speaking, the formation process of the third type as mentioned above pores is divided into three steps[15]: keyhole fluctuations leading to bubbles formation; bubbles floating to the back of the molten pool; and bubbles being captured by the solidification front. Noting that destroying any of these steps can

reduce or eliminate pores. Furthermore, the number of pores depends not only on the number of bubbles, but also on the solidification rate. Therefore, a larger molten pool contributes to the migration of bubbles[16]. The formation of pore in aluminum alloy laser welding is closely related to the stability of the keyhole[17] and the volatility of the molten pool[18,19], and the violent molten pool flow around the keyhole leads to instability of the keyhole, making the possibility of pore escape less likely. In laser welding of aluminum alloy T-joints, Dittrich [20] pointed out that the least number of pore defects was found when the keyhole was symmetrically distributed, but did not reveal the reason for this. Yang's [9] simulation results further revealed that reducing the weld heat input led to the gradual weakening of the keyhole connection, and even formed two separate keyholes. The convection above the keyhole turns into downward flow, vortices form in the cross section and the molten pool flow becomes more complex, which greatly reduces the efficiency of bubbles overflow and forms pore defects in the weld.

In terms of laser welding porosity suppression, some scholars[21] have achieved double-sided forming without pore defects in AA2024-AA7050 heterogeneous aluminum alloys by reasonably adjusting the welding process parameters to increase the keyhole size and uniform molten pool temperature distribution, thus improving the escape conditions of bubbles. Others scholars[11] have also reduced metallurgical porosity by controlling the welding heat input to reduce the volatilization of low boiling point elements during wire-free filler welding in T-joints; using a faster welding speed (about 2.5m/min) to shorten the solidification time of the molten pool and prevent pore aggregation. In summary, the strategy of suppressing porosity through the adjustment of process parameters has major limitations. The most effective method is the use of an oscillating laser beam, which not only creates welded joints without obvious cracks and porosity defects[22,23], but also improves molten pool flow and homogenizes the weld alloying elements thus alleviating macroscopic segregation [24]. However, considering the experimental conditions and costs, it is difficult to directly observe experimentally how the oscillating beam affects the molten pool flow and keyhole evolution, and influences the formation of porosity in the weld.

With the development of computer technology, numerical models are increasingly used to simulate the laser welding process to study the melt flow and the evolution of keyhole to reveal the mechanism of pore formation[25], [26], [27]. The use of simulation has become an essential step in the closed loop of experiment-simulation interpretation-proposed regulation method-experimental verification. The basic numerical models are based on the governing equations and the boundary conditions set according to the situation to work out the temperature and velocity fields of the model[28]. The credibility of the model was verified by comparing the experimental and simulated melting zone profiles[29,30] or the degree of agreement of the temperature field[31]. A numerical model that matches well with experiments can accurately predict weld depth and width[32], and can also help explain the mechanisms of phenomena that are difficult to observe directly in experiments[33]. In order to

make the numerical model more accurate, appropriate algorithms and physical models are needed: the volume of fluid (VOF) method is indispensable in tracking the dynamic evolution of the free interface[34]. Surface tension[35] and recoil pressure[15] contribute significantly to molten pool fluctuations and keyhole generation, respectively. In addition, for laser deep penetration welding, Fresnel absorption and multiple reflection models[36] tend to provide more reliable simulation results; adiabatic bubble models[16] are employed to investigate the mechanisms of pore generation. In analyzing the effect of oscillating laser beam on the material, it is common to focus on a beam oscillation period[34,37], and the energy density distribution of the oscillating laser beam is also essential[38]. In the numerical simulation of wire-fill welding of T-joints, the tabs[39] and dripping metal droplets[38] are used as the fused-in wire, however, this simplification may make the numerical model less accurate.

Therefore, in this work, the dual-laser beam oscillating bilateral synchronous welding (DLBOW) process was proposed for the first time and successfully reduced the porosity in T-joints. Secondly, a numerical CFD model with a complete wire melting process was established in agreement with the experimental results. The keyhole morphology was traced by the volume of fluid method (VOF) and ray tracing model. Also, the Fresnel multiple reflection model was used to restore the evolution of the keyhole during the welding process. In addition, the influence of the molten pool flow behavior on the weld seam morphology during the welding process of DLBOW and DLBW processes was studied. The keyhole evolution behavior of DLBOW under a beam motion cycle with different beam incidence angles was compare. Finally, the forces on the keyhole walls at different laser incidence angles are fully discussed using a quasi-steady state model, revealing the keyhole evolution regulation and the mechanism of pore formation at different laser incidence angles.

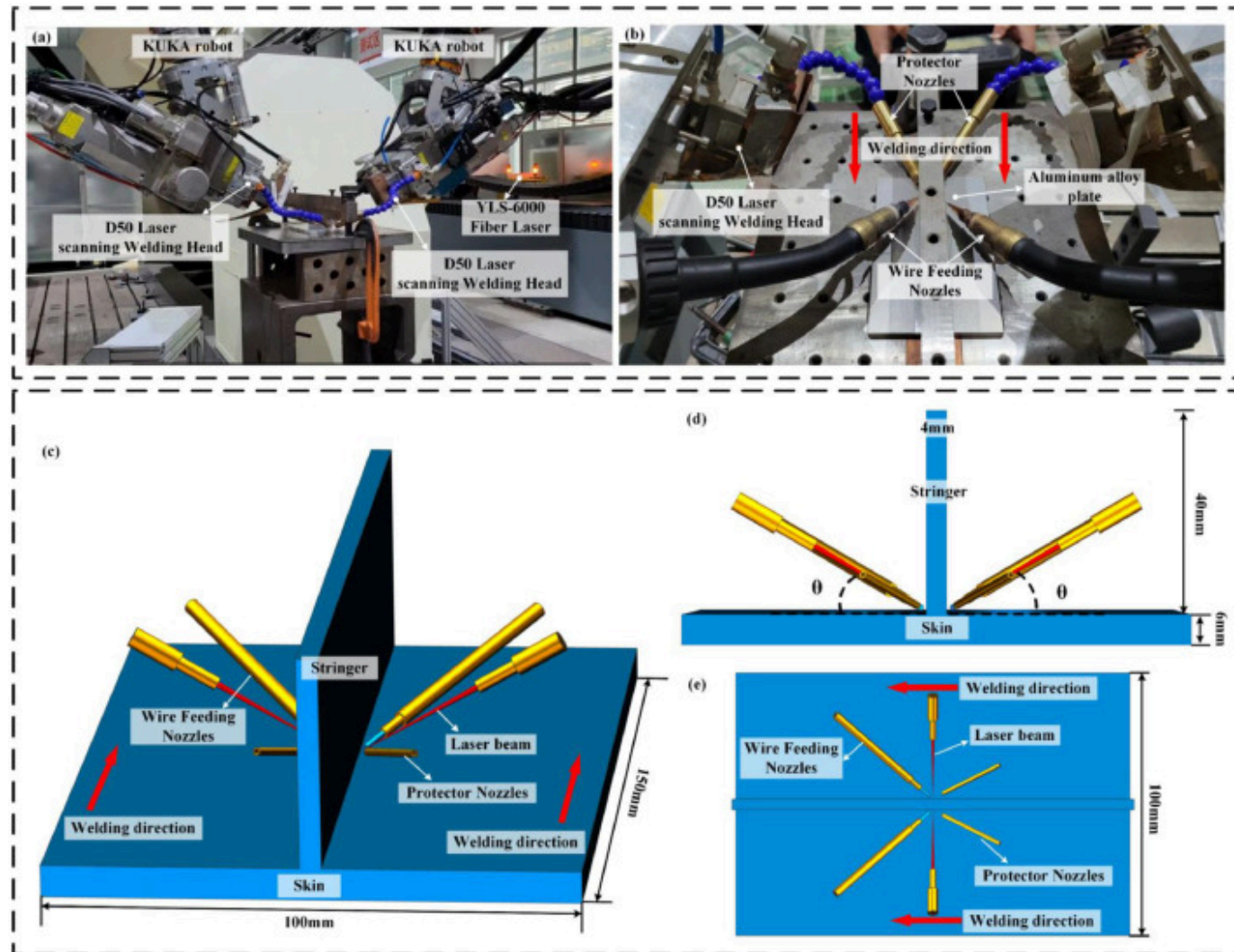
2. Experimental details

In this work, a plate of size 150mm×100mm×6mm is used as the skin and the stringer size is 100mm×40mm×4mm. ER2319 wire with a diameter of 1.2mm was used. The chemical composition of the plate and welding wire is shown in [Table 1](#). The experimental equipment shown in [Fig. 1](#) consists of a YLS-6000 fiber laser, two D50 laser scanning welding heads and a six-axis KuKa robot. The fiber diameter is 200μm and the spot diameter is 400μm. The amplitude distribution of the beam cross section obeys a Gaussian function. As shown in [Fig. 1\(b\)](#), the laser heads, wire feeding heads and protector nozzles on both sides of the stringer are arranged symmetrically. As in [Figure 1\(d\)](#), the plane where the lasers are located is perpendicular to the skin, and the angle between the beam with the skin is θ . The wire feeding heads and protector nozzles are located in the same inclined plane as the laser beam. The process parameters used in the experiment are shown in [Table 2](#). In order to ensure that the weld formation is as similar as possible in both processes. In our experiments the laser power of DLBW is lower than that of DLBOW.

After the experiments were completed, 3D X-ray computed tomography (CT) scanning analysis methods were used to obtain the pore distribution characteristics of the T-joint welds and calculate the porosity.

Table 1. Chemical composition of 2219 aluminum alloy and ER2319 welding wire used in this experiment.

Material	Mn	Si	Ti	Zr	V	Cu	Mg	Fe	Zn	Al
BM	0.2~0.4	≤0.2	0.1~0.2	0.1~0.2	0.05~0.15	5.8~6.8	0.2~0.4	≤0.3	≤0.1	Balanced
wire	0.2~0.4	≤0.2	0.02~0.1	0.1~0.25	0.05~0.15	5.8~6.8	≤0.02	≤0.3	≤0.1	Balanced



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Fig. 1. Schematic diagram of DLBW/DLBOW:(a) the main view of laser welding experiment (b) two laser heads (c) the three-dimensional schematic diagram; (d) section view; (e) overhead view

Table 2. Details of process parameters researched in this experiment.

	DLBOW			DLBW		
	3900	3900	3900	3500	3500	3500
Laser power P (W)						
Welding velocity V(m/min)	1.2	1.2	1.2	1.2	1.2	1.2
Wire feed rate V_w (m/min)	5	5	5	5	5	5
Scanning amplitude A(mm)	1.0	1.0	1.0			
Scanning frequency f (Hz)	300	300	300			
Incident angle θ (°)	20	30	40	20	30	40

3. Mathematical models

The heat transfer, keyhole evolution and molten pool flow processes during laser welding are very complex. In order to improve the computational efficiency without compromising the accuracy of the model, the following assumptions need to be made about the model:

- 1 The flow is laminar and the liquid aluminum in the molten pool is considered to be an incompressible Newtonian fluid
- 2 The metal is assumed to be isotropic and homogeneous
- 3 The initial temperature of the T-joint is set to 293 K. The effect of plasma on the welding process is negligible

3.1. Heat source in CFD model

The transverse distribution of the light field is constant at any position along the optical axis propagation direction in the stable optical cavity, but the beam diameter size is variable. The effective radius $w(z)$ of the TEM₀₀-mode Gaussian beam in the

diffraction limit varies along the z-direction of the cavity axis with a hyperbolic law as follows [40]:

$$\frac{w^2(z)}{w_0^2} - \left(\frac{\lambda z}{\pi w_0^2} \right)^2 = 1 \quad (1)$$

where, λ is the laser wavelength, w_0 is the spot radius at the beam focus. The distance from the beam waist to the cross-section where the beam area is twice the beam area at the beam waist is the Rayleigh length z_R . Calculated from [equation\(1\)](#) $z_R = \frac{\pi w_0^2}{\lambda}$. The light intensity of a TEM₀₀ mode beam propagating in the z-direction can be expressed as [41]:

$$I_{x,y,z} = (1 - \eta) \cdot \frac{3Q}{\pi w^2(z)} e^{\left[-\frac{3(x^2+y^2)}{w^2(z)} \right]} \quad (2)$$

Where, η indicates the power attenuation coefficient, takes the value of 40%, Q is the laser power. "x", "y", "z" are the coordinates of a point in a three-dimensional coordinate system.

3.2. Governing equations

In the Cartesian coordinate system, the temperature and velocity fields of the fluid are obtained by solving the governing equations for the fluid flow dynamics of the welding process [39,42]:

Mass conservation equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \vec{v} = 0 \quad (3)$$

Where, ρ is fluid density, t is time, $\vec{v}$ is velocity vector.

Momentum conservation equations:

$$\rho \frac{\partial \vec{v}}{\partial t} + \rho \nabla \cdot (\vec{v} \otimes \vec{v}) = -\nabla P + \mu \nabla^2 \vec{v} + \vec{f}_b - \frac{\mu \vec{v}}{K} \quad (4)$$

Where, P is ambient pressure, μ is dynamic viscosity, K is permeability of porous media.

Energy conservation equation:

$$\rho \frac{\partial H}{\partial t} + \rho \nabla \cdot (\vec{v} H) = \nabla \cdot (k \nabla T) + q \quad (5)$$

Where, H is enthalpy, k is thermal conductivity of the material, T is the temperature, q is the external source terms of energy.

Using Boussinesq model to deal with thermal buoyancy caused by temperature gradient. Thermal buoyancy f_b in the same direction as the acceleration of gravity. Therefore, it is only added to the z-direction momentum conservation equation:

$$f_b = \rho_0 \beta \vec{g} (T - T_{ref}) \quad (6)$$

Where, T_{ref} is the ambient temperature, ρ_0 is the density of the fluid at T_0 , β is the coefficient of thermal expansion, and $\vec{g}$ is the acceleration of gravity.

3.3. Solidification and phase change model

During the solidification of the molten pool, there is a certain momentum loss of the flowing liquid molten pool metal due to the gradual increase of the solid phase fraction in the transition region. The solid-liquid interface is treated by enthalpy porosity technology, which treats the mushy zone as a porous medium with isotropic permeability[43]. Porosity of each control volume is the liquid phase volume fraction.

The permeability of porous media K can be obtained by the Karman-kozeny equation:

$$K = \frac{f_l^3 + \delta_0}{A_m (1 - f_l)^2} \quad (7)$$

Where, δ_0 is a non-zero fractional denominator constant. A_m is the Darcy constant in the mushy zone. The expression for the liquid phase volume fraction f_l [44] is as follows:

$$f_l = \begin{cases} 0 & T < T_s \\ \frac{T - T_s}{T_l - T_s} & T_s < T < T_l \\ 1 & T > T_l \end{cases} \quad (8)$$

Where, T_s and T_l are the solidus and liquidus temperature. In the solid phase region, $f_l = 0$; in the liquid phase region, $f_l = 1$; and in the mushy region, $0 < f_l < 1$.

The volume-of-fluid (VOF) method is used to trace the free surface of the molten pool by solving the continuity equation for the phase volume fraction [45]:

$$\frac{\partial f_{VOF}}{\partial t} + \nabla \cdot (v f_{VOF}) = 0 \quad (9)$$

Where f_{VOF} is the phase volume fraction of the controled volume. When $f_{VOF}=0$, the control volume is free of metallic liquids. If $f_{VOF}=1$, the control volume is occupied by a metallic liquid. If $0 < f_{VOF} < 1$, it means that the surface of the liquid is located inside the control volume.

3.4. Multi-reflection mode

When the welding process forms the keyhole, the laser beam will reflect inside it several times, increasing the energy density at the bottom of the keyhole, making it become deeper and sharper. The CFD model uses a multiple reflection model with Fresnel absorption to study the effect of keyhole evolution and molten pool flow on pore formation during double-beam laser welding of T-joints.

The reflected ray x_r can be expressed as [36,46]:

$$x_r = x_i - 2(x_i \cdot n)n \quad (10)$$

x_i is the incident light, n is the incident point normal vector, in multiple reflections, the incident light will be reflected in the keyhole several times until its energy is absorbed or reflected from the keyhole.

In Fresnel absorption based laser deep penetration welding, the absorption rate is expressed by the following equation [40,47]:

$$a = 1 - \frac{1}{2} \left(\frac{1+(1-\varepsilon\cos\alpha)^2}{1+(1+\varepsilon\cos\alpha)^2} + \frac{\varepsilon^2-2\varepsilon\cos\alpha+2\cos^2\alpha}{\varepsilon^2+2\varepsilon\cos\alpha+2\cos^2\alpha} \right) \quad (11)$$

where α is the angle between the incident light and the surface normal vector n , and ε is a coefficient that depends on the laser

beam type and material properties[48]:

$$\varepsilon^2 = \frac{2\varepsilon_2}{\varepsilon_1 + \left[\varepsilon_1^2 + \left(\frac{\sigma_{st}}{\omega\varepsilon_0} \right)^2 \right]^{\frac{1}{2}}} \quad (12)$$

Where, ε_{st} is the electric conductivity depth per unit (base material), ε_2 is relative permittivity (base material), ε_1 is relative permittivity(plasma), ε_0 is permittivity of vacuum, ω is laser regular frequency. The value of ε for this model is taken as 0.10. In our numerical model, the absorbed energy is equal to the energy of the incident beam multiplied by the absorption rate.

3.5. Driving force

During the welding process, the formation and evolution of the keyhole and the molten pool flow behavior are influenced by the recoil pressure P_R , surface tension P_s , thermal buoyancy f_b , protective gas dynamic pressure P_D and gravity.

During the welding process, laser power density of deep fusion welding is high(power density $>10^6 \text{W/cm}^2$), the metal fluid at the molten pool interface will absorb heat and evaporate, and the resulting metal vapor will apply reverse pressure to the molten pool, which is called recoil pressure P_R , and it can promote molten pool flow and keyhole formation[49]:

$$P_R(T) = \eta P_{sat} = \eta P_v \exp \left\{ \frac{\Delta H_v}{(\tau-1)c_v T_v} \left(1 - \frac{T_v}{T} \right) \right\} \quad (13)$$

where, η is evaporation coefficient, $0 \leq \eta \leq 1$; P_{sat} is evaporation pressure, P_v is saturation pressure, ΔH_v is evaporation latent heat, τ is the ratio of the vapor specific heats, c_v is steam constant specific heat and T_v is saturation temperature.

Surface tension P_s is a function of temperature and has an important effect on the molten pool flow during the welding process:

$$P_s = \gamma + A_\gamma (T - T_l) \quad (14)$$

where, γ represents the coefficient of surface tension A_γ represents the temperature gradient of surface tension.

In the laser oscillating welding process, the effect of thermal buoyancy f_b on the liquid metal due to the temperature gradient in the molten pool cannot be neglected and is expressed in Eq.(6).

Protective gas is applied above the molten pool during laser oscillation welding, which in this CFD model is achieved by applying dynamic pressure on the molten pool surface, protective gas dynamic pressure P_D :

$$P_D = \frac{1}{2} \rho v^2 \cos \alpha_0 \quad (15)$$

where, ρ is the gas density, v is the gas velocity and α_0 is the angle between the surface normal vector of the small hole with the gas velocity vector.

3.6. Coordinate system transformation

As shown in [Figure 2](#), the welding direction is along the positive x-axis, and the two laser beams are at an angle of θ with the skin. The coordinate system is transformed accordingly, and the light sources on the left and right need to be rotated counterclockwise and clockwise by an angle θ . The formula for calculating the position of the laser beams at any moment can be expressed as:

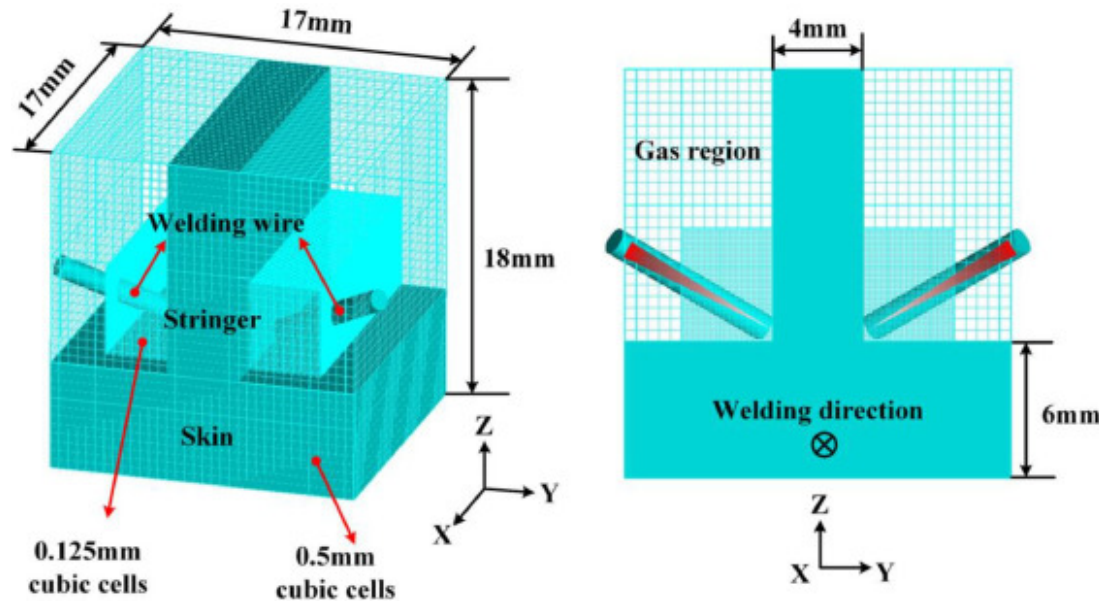
$$\begin{cases} x_l = \frac{A}{2} \sin(2\pi ft) \\ y_l = \frac{A}{2} \sin(\theta) \cos(2\pi ft) \\ z_l = \frac{A}{2} \cos(\theta) \cos(2\pi ft) \end{cases} \quad (16)$$

$$\begin{cases} x_r = \frac{A}{2} \sin(2\pi ft) \\ y_r = \frac{A}{2} \sin(\theta) \cos(2\pi ft) \\ z_r = \frac{A}{2} \cos(\theta) \cos(2\pi ft) \end{cases} \quad (17)$$

where, x_l , y_l , z_l , x_r , y_r , z_r correspond to the left and right axes respectively, the laser oscillation amplitude is A and the frequency is f. The laser velocity equation is obtained by deriving the above equation:

$$\begin{cases} v_{xl} = A\pi f \cos(2\pi ft) \\ v_{yl} = A\pi f \sin(\theta) \sin(2\pi ft) \\ v_{zl} = -A\pi f \cos(\theta) \sin(2\pi ft) \end{cases} \quad (18)$$

$$\begin{cases} v_{xr} = A\pi f \cos(2\pi ft) \\ v_{yr} = -A\pi f \sin(\theta) \sin(2\pi ft) \\ v_{zr} = -A\pi f \cos(\theta) \sin(2\pi ft) \end{cases} \quad (19)$$



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Fig. 2. The schematic of the simulation domain

The above equations v_{xl} , v_{yl} , v_{zl} , v_{xr} , v_{yr} , v_{zr} correspond to the left and right laser velocity components respectively.

3.7. Boundary conditions and simulation cases

Thermal boundary conditions for T-joints [50]:

$$k \frac{\partial T}{\partial n} = -h_c (T - T_{ref}) - \epsilon_r \sigma (T^4 - T_{ref}^4) \quad (20)$$

where, $\mathbf{n}$ is the normal vector at the free interface, h_c is the convective heat transfer coefficient, ϵ_r the thermal emissivity, σ is the Stephan-Boltzmann constant, T_{ref} is the ambient temperature.

Free surface of the keyhole [45]:

$$k \frac{\partial T}{\partial n} = q_{abs} - \varepsilon_r \sigma (T^4 - T_{ref}^4) - h_c (T - T_{ref}) - W_{evp} H_v \quad (21)$$

where, q_{abs} is the absorbed laser energy, W_{evp} and H_v are the evaporation rate and the latent heat of evaporation of the liquid metal.

The thermal physical parameters of the 2219 aluminum alloy are shown in [Table 3](#), and the schematic diagram of the CFD numerical model is shown in [Fig 2](#). The model size of the whole calculation domain is 17mm×17mm×18mm. Considering the computational efficiency and simulation accuracy, the overall model is divided into cubes with side length of 0.5mm, and the finer grid of 0.125mm is used for the weld area. The skin (0<z<6mm), stringer (6mm<z, -2mm<y<2mm) and the welding wire on both sides of the stringer constitute the solid part of the model, and the rest of the model is air domain. In our model, an inclined cylindrical solid fluid is set up as a welding wire from each side of y_{min} and y_{max} , given that the fluid has the same velocity as the wire. The geometric angle relationship between the laser beam, the welding wire and the plate is strictly according to the experimental setup, and the laser beam on each side moves in a circular motion with its own axis. The initial time step for the calculation is 2e-5s and the minimum time step is 1e-15s. Eventually, the CFD model was solved by Flow-3D-v.12.

Table 3. Thermal physical properties of 2219 aluminum alloy.

Name	Symbol	Unit	Value
Solidus density	ρ_s	kg m ⁻³	2720
Liquidus density	ρ_l	kg m ⁻³	2520
Thermal conductivity of solid	k_s	W m ⁻¹ K ⁻¹	236
Thermal conductivity of liquid	k_l	W m ⁻¹ K ⁻¹	120
Specific heat capacity of solid	C_{ps}	J kg ⁻¹ K ⁻¹	860
Specific heat capacity of liquid	C_{pl}	J kg ⁻¹ K ⁻¹	1140
Latent heat of fusion	H_m	J kg ⁻¹	3.97×10 ⁵

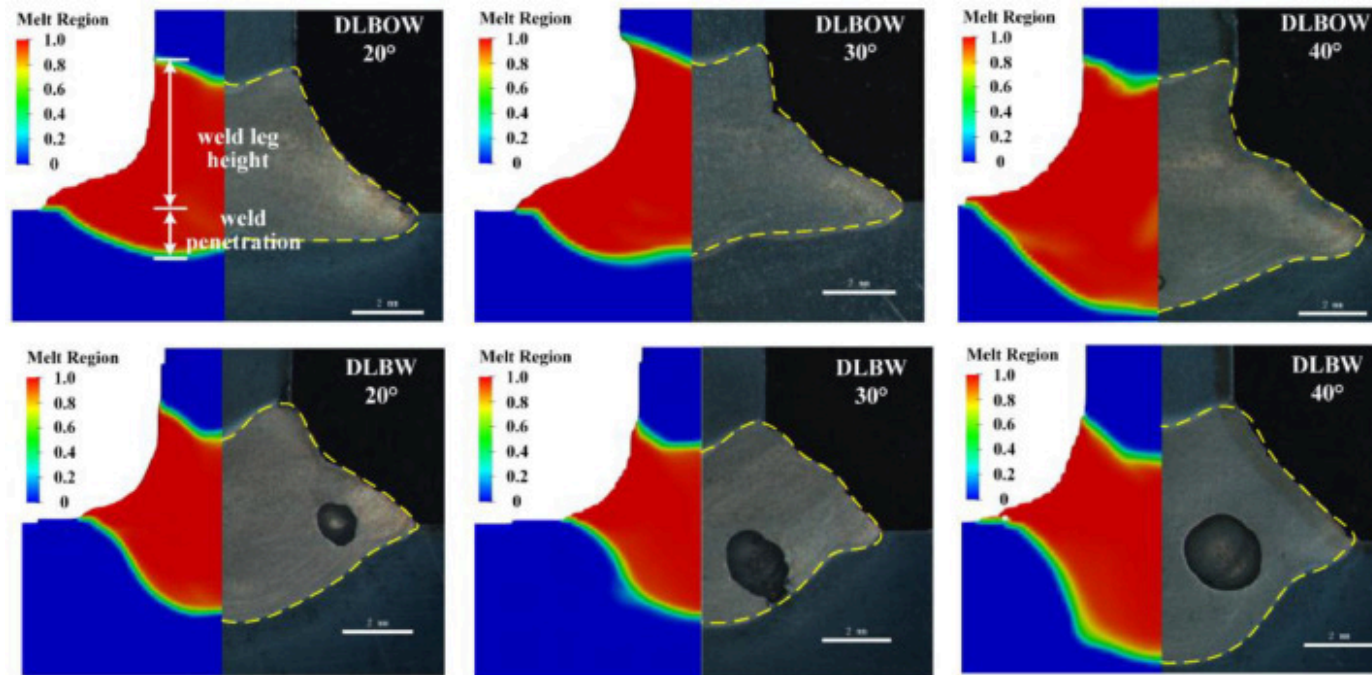
Name	Symbol	Unit	Value
Latent heat of evaporation	H_v	J kg ⁻¹	1.09×10^7
Coefficient of surface tension	γ	N m ⁻¹	0.998
Temperature gradient of surface tension	A_δ	N m ⁻¹ K ⁻¹	-0.33×10^{-3}
Dynamic viscosity	μ	Pa s	0.015
Coefficient of thermal expansion	β		3.12×10^{-5}
Solidus temperature	T_s	K	820
Liquidus temperature	T_l	K	916
Convective heat transfer coefficient	h_c	W K ⁻¹ m ⁻²	20
Thermal emissivity	ε_r		0.9
Ambient temperature	T_{ref}	K	293

4. Results and discussion

4.1. Reliability verification of model

As shown in Fig.3, in order to verify the reliability of the model, the experimental and simulated weld cross-sectional morphology was compared under different oscillation parameters. The simulation results are located on the left, with the red part indicating the melted area and the weld profile on the right depicted by the yellow dashed line. Table4 shows the experimental and simulated weld cross-sectional characteristic dimensions weld penetration (W_p) and weld leg height (W_h) for different oscillating parameters as well as the respective porosity under CT. Based on the experimental data, it can be found that W_p increases from 1.45mm to 2.32mm and W_h increases from 3.91mm to 4.63mm with increasing laser incidence angle under DLBOW, and the simulation results also demonstrates this tendency. The characteristic dimensions of the weld seam for DLBW 20° and DLBW 30° are relatively close, with W_p and W_h around 2.9 mm and 3.0 mm, respectively. While the W_p and W_h of DLBW

40° are 3.39mm and 3.79mm. The simulated cross-sectional weld dimensions correspond well with the experiments, verifying the reliability of the mode. The data of the porosity calculated from the CT scan shows that the porosity is dramatically reduced under the DLBOW process. Meanwhile, the porosity is further reduced as the laser incidence angle decreases. It indicates that the laser oscillation and small incidence angle can promote the reduction of porosity.



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Fig. 3. Comparison of experimental and simulated cross-sections at different process parameters

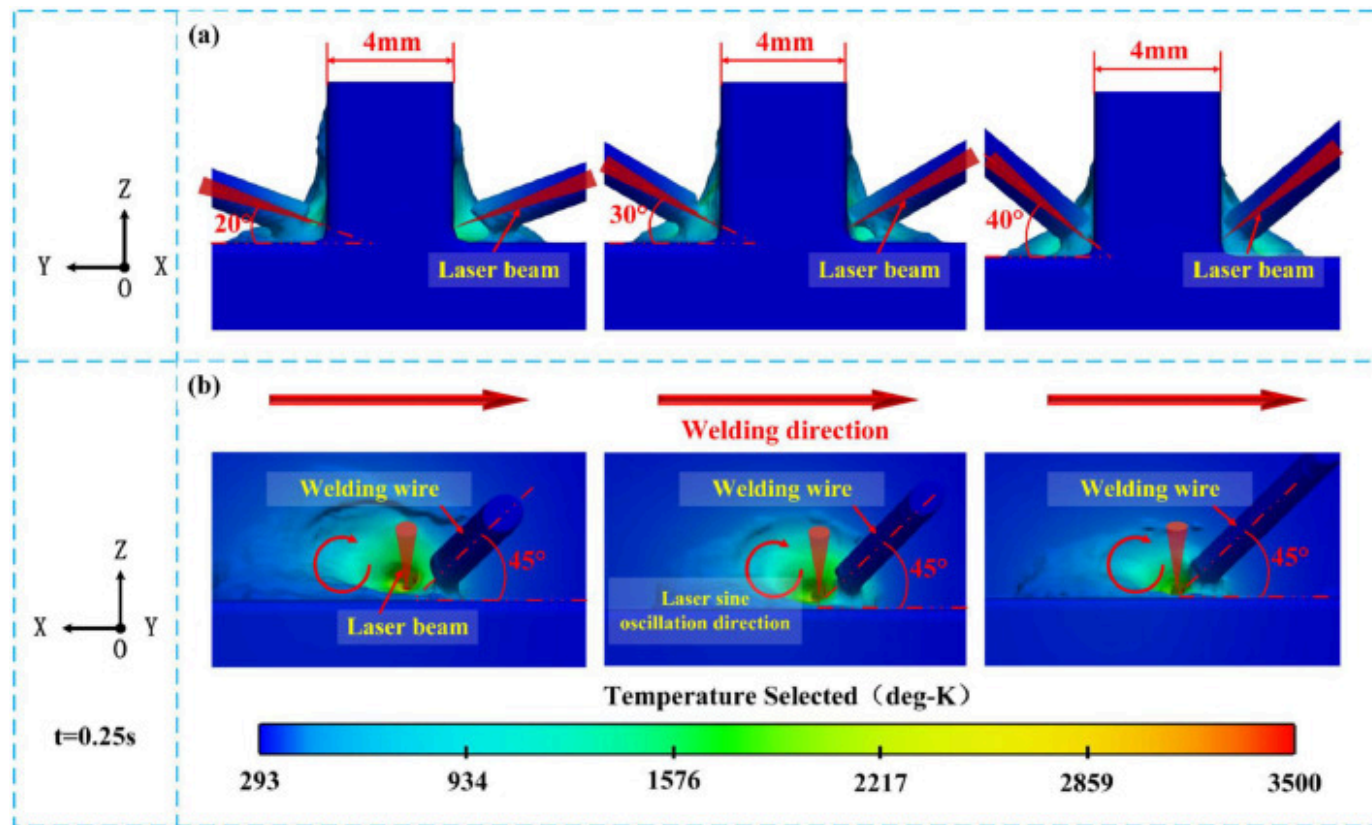
Table 4. Comparison of the characteristic dimensions of T-joint welds under experiment and simulation and porosity of different parameters.

	Weld penetration(W_p , mm)			Weld leg height(W_h , mm)			Porosity(%)
	Experiment	Simulation	Precision	Experiment	Simulation	Precision	
DLBOW20°	1.45	1.40	96.6%	3.91	3.80	97.3%	0.26
DLBOW30°	1.68	1.61	95.8%	4.21	4.13	98.1%	0.43
DLBOW40°	2.32	2.25	97.0%	4.63	4.14	89.4%	0.46
DLBW20°	2.91	2.85	97.9%	3.04	3.45	88.1%	3.45
DLBW30°	2.82	2.75	97.5%	2.99	3.35	89.3%	4.07
DLBW40°	3.39	3.27	96.5%	3.97	3.24	81.6%	4.56

4.2. Molten pool flow behavior

4.2.1. Melting behavior of welding wire

Fig.4(a) shows that the laser beam is located in the YOZ plane and its angle with the skin is θ . The angle of the welding wire with the skin in the YOZ plane is consistent with the laser incidence angle. The angle between the wire and the horizontal plane in the XOZ plane is kept constant at 45° in Fig.4(b). From Fig.4(b), it can be observed that as the laser incidence angle increases, the welding wire gradually deviates from the plane where the laser beam is located. The position of the laser-wire interaction is biased downward and the length of the welding wire melted by the laser beam is shortened. At the laser incidence angle of 20° , the energy of the laser beam acts directly on the welding wire. The transition point between the wire and the plate is located on the skin side. At a laser incidence angle of 40° , the energy of the laser beam acts more on the molten pool. The wire enters the molten pool to absorb heat and melt. Front feeding conditions, filler wire melted into the molten pool flow direction opposite to the liquid molten pool, reducing the stability of the molten pool and keyhole, increasing the possibility of keyhole closure, which is not conducive to the escape of air bubbles and increases the chances of pore defects. Since the DLBOW40° wire is in the form of liquid bridge transition, the impact effect of the liquid wire melted into the molten pool is more obvious. Therefore, DLBOW20° keyhole stability is higher, which helps the escape of bubbles and reduces the weld pore defect.



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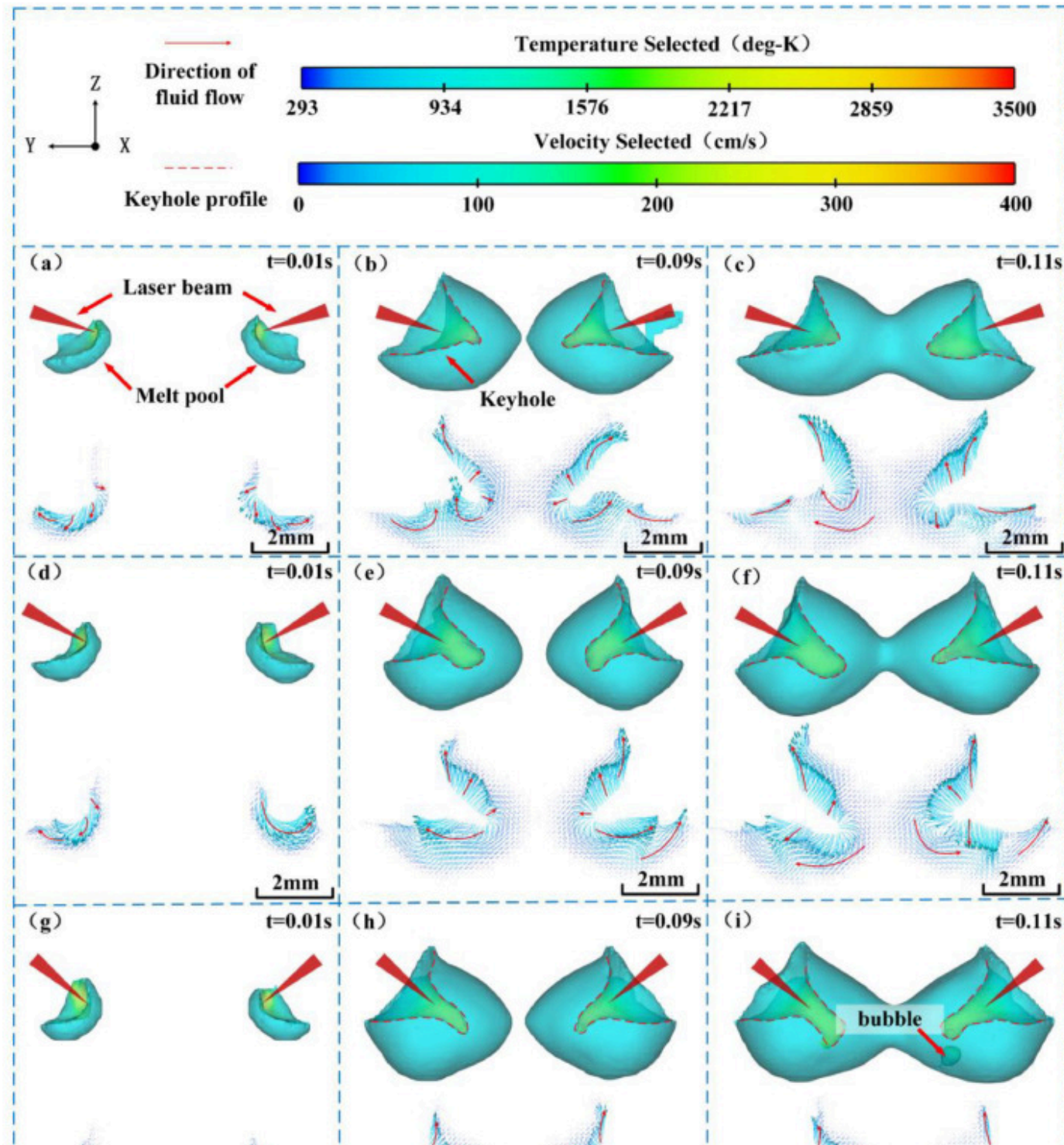
Fig. 4. Weld wire transition at different process parameters:(a) Cross-sectional view (b) XOZ Viewpoint

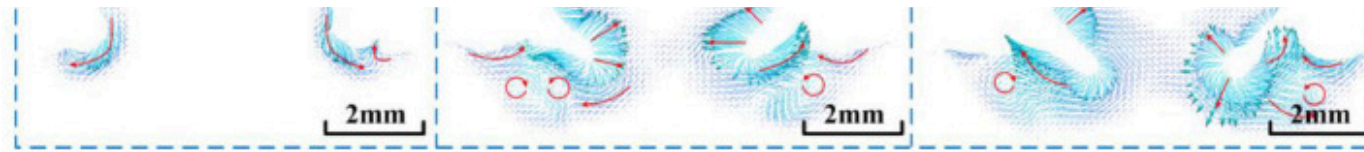
4.2.2. Flow behavior of the molten pool expansion phase at different angles of DLBOW

Fig.5 shows the temperature field and its corresponding velocity vector field in the expansion period of the molten pool for different laser incidence angles in YOZ view. At $t=0.01s$, for DLBOW20°, the skin is partially melted at the connection with the stringer and the molten pools on both sides of the stringer resemble a "C" shape. Restricted by the geometric form of the skin and stringer joint, the metal liquid flow is outward along the skin. At this time, the distance between the two sides of the molten pool is relatively far. At $t=0.09s$, there are obvious keyholes in the molten pool on both sides of the stringer, and the distance between

the molten pools is reduced and the size is significantly enlarged. At an incidence angle of 30° , the evolution of the molten pool at the three moments is similar to that at 20° , but the keyhole depth is slightly larger. When the laser incidence angle is 40° , the bottom of the keyhole is sharper, the vortex appearing at the bottom of the molten pool is more obvious, and bubbles also appear in the molten pool at $t=0.11\text{s}$. Three kinds of liquid flow forms are found in the velocity vector field according to Fig.5:

- (1) The flow of liquid along the normal direction of the keyhole wall, towards the interior of the molten pool, and this flow leads to the expansion of the keyhole.
- (2) The flow of the molten pool expanding outward: upward along the stringer wall, this flow influences the magnitude of the W_h ; outward along the skin, this flow influences the magnitude of the W_p ;
- (3) The flow of metal liquid to the outer edge of the skin, back into the molten pool as well as the inflow of melted wire, this flow is opposite to the flow of flow (2), causing shock and instability of the molten pool.





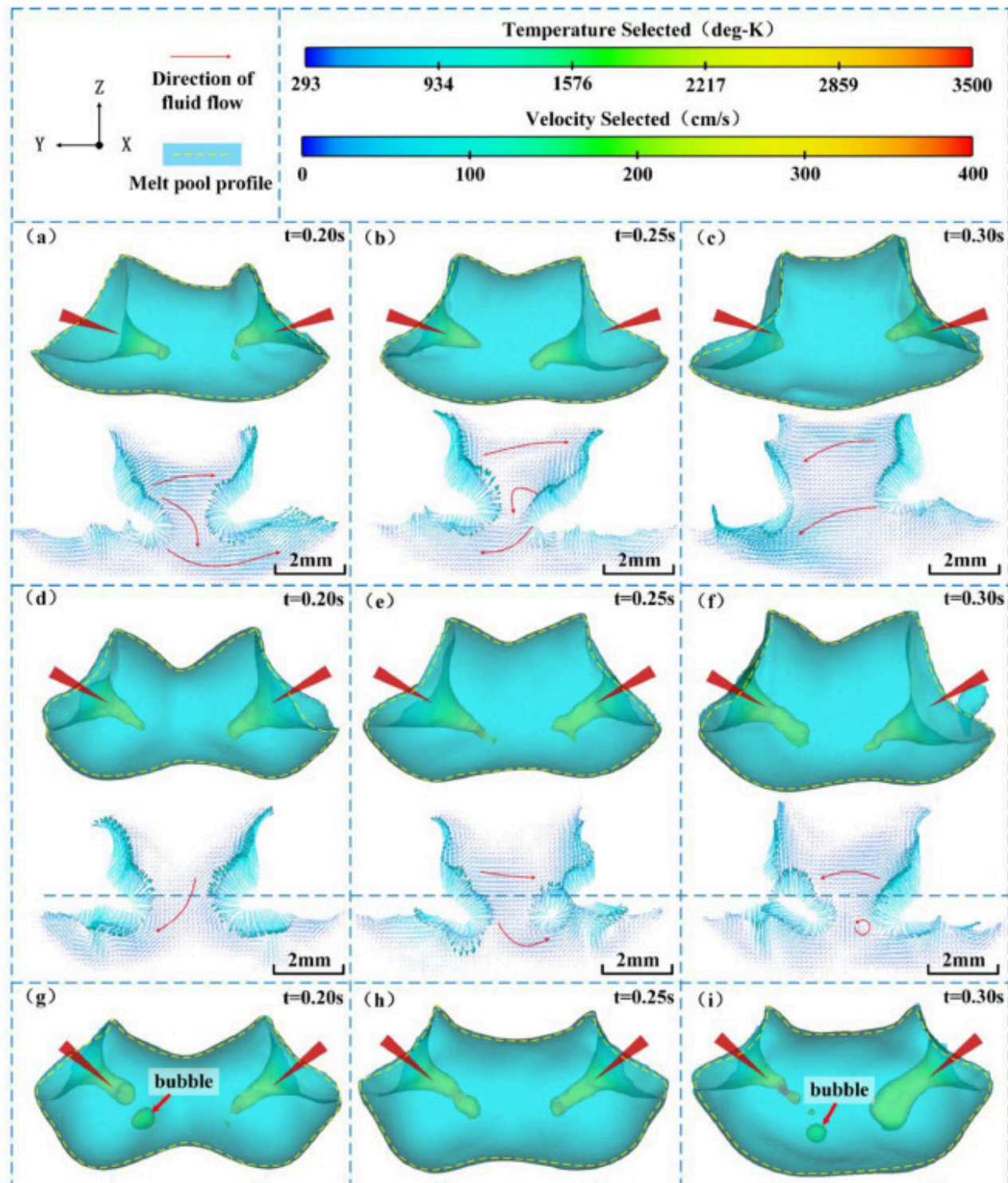
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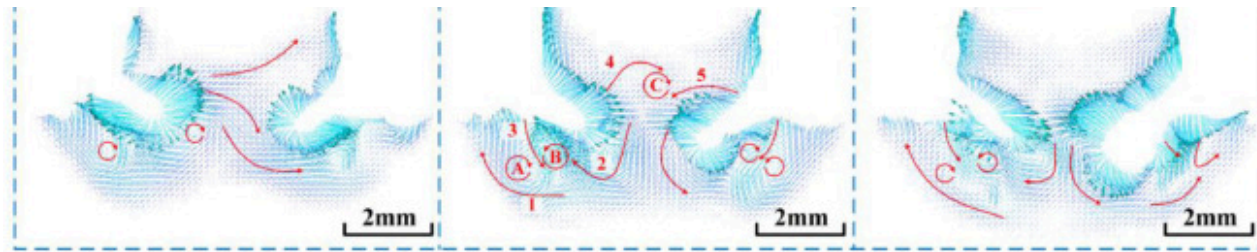
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Fig. 5. Temperature and velocity vector fields during the evolution of the molten pool at different laser incidence angles of DLBOW at the initial moment (a) ~ (c) :DLBOW20°; (d) ~ (f) :DLBOW 30°; (g) ~ (i) :DLBOW 40°

In summary, at the initial moment, both sides of the plate melt under the effect of the laser beam to form a molten pool respectively. The molten pool expands continuously and connects between 0.09s and 0.11s. The tilt of the keyhole is highly consistent with the angle of incidence of the laser beam. With the increasing angle of laser incidence, the flow of the molten pool becomes more intense and the depth of the keyhole increases slightly.

As shown in Fig.6, when the laser incidence angle is 20° and 30°, the profile above the molten pool is "V" shaped, and with the time from 0.20s to 0.30s, the bottom profile of the molten pool is from "W" shaped to approximately "U" shape. For DLBOW40°, the profile above the molten pool evolves from a "V" to a "U" shape, and the profile at the bottom of the molten pool evolves from a "W" to a "U" shape. As the laser incidence angle increases, the molten pool depth is significantly larger, which is due to the larger vertical component of the laser at large angles.





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Fig. 6. Temperature and velocity vector fields during molten pool evolution for different laser incidence angles at DLBOW when the molten pool is connected. (a) ~ (c) :DLBOW 20°; (d) ~ (f) :DLBOW 30°; (g) ~ (i) :DLBOW 40°

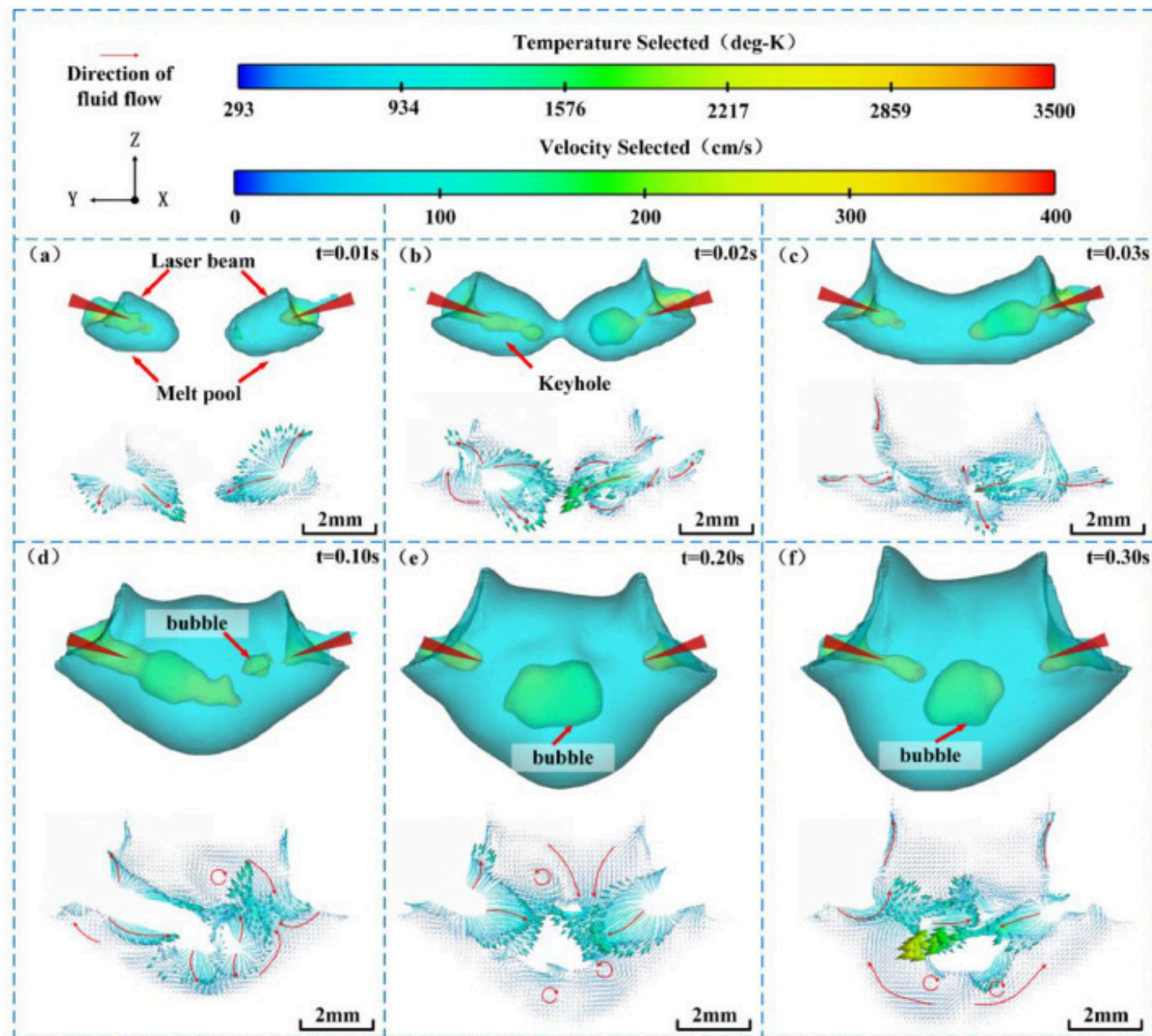
The liquid flow between the two sides of the molten pool is marked in the velocity vector fields in Fig.6 (a~f). The liquid flow is relatively consistent from one side of the molten pool to the other side, and the overall flow is very smooth, with only a relatively small vortex found in (f). This means that the flow of liquid in this part is consistent and smooth, unlike a vortex which can cause bubbles to spin around in it; in addition, the flow from one side to the other means that bubbles can escape from the opposite side of the molten pool. Therefore, this phenomenon is beneficial for the escape of bubbles from the molten pool during the welding process. Fig.6(g~i) show the flow behavior of the molten pool at an incidence angle of 40°, the flow between the molten pools is weakened. As shown in flow 1 in the Fig.6(h), the liquid flow at the bottom of the molten pool is obstructed by the skin and flows upwards along the bottom. The flow in the opposite direction, like flow 1 and flow 3, tends to lead to the formation of vortices.

With the increase of laser incidence angle, the size of the molten pool increases slightly, the flow becomes more turbulent and the depth of the keyhole increases, which is caused by the larger actual heat input to the molten pool at large angles. "actual heat

input" is the actual laser energy absorbed by the molten pool. At an angle of 40° , the more elongated keyhole tends to collapse under the impact of liquid flow to form bubbles, further confirming that the molten pool is more turbulent at large angles, the keyhole evolution process is unstable, and the tendency of pore formation is greater.

4.2.3. Comparison of molten pool flow behavior between DLBOW and DLBW

As shown in the Fig.7, at DLBW 20° , the molten pool expands sharply on both sides under the effect of the laser beam. Between 0.01s~0.02s both sides of the molten pool are connected and the bottom profile of the molten pool evolves rapidly from "W" to "U" shape, and then to "V" shape. The typical shape of the bottom profile for a T-joint molten pool reflects the actual heat input to the molten pool. The shape of DLBW 20° keyhole is quite irregular and the molten pool flow is extremely violent. Under the impact of the metal flow, the keyhole is susceptible to collapse and form bubbles. Moreover, the bubbles formed in Fig.7(e~f) are located in the middle and bottom of the molten pool, the escape path is far away and it is prone to form pore defects.

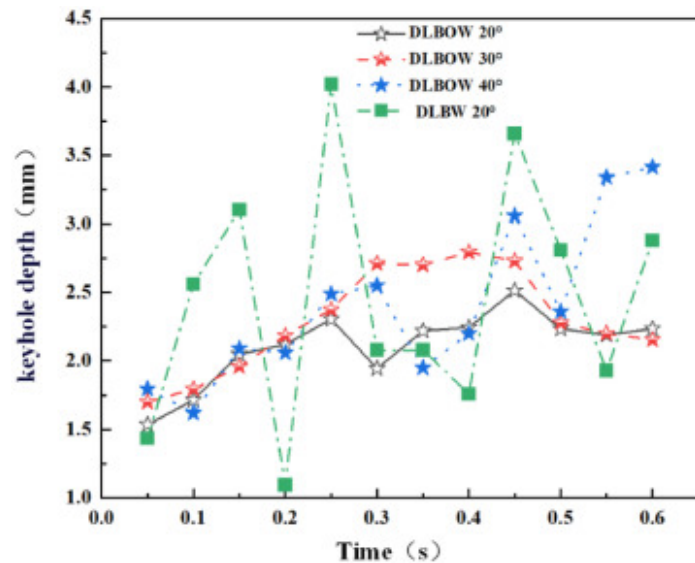


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Fig. 7. Temperature and velocity vector fields during molten pool evolution at DLBW20°

In order to quantify the variation of keyhole depth, the average depths of keyholes on the left and right sides at different process parameters were counted at 0.05s intervals in Fig.8. In the DLBOW process, the average keyhole depths at 20°, 30° and 40° were 2.08 mm, 2.30 mm and 2.41 mm, respectively, with increasing laser incidence angles, the keyhole depth increases, and this trend was also observed for each individual moment. It should be noted that the three moments of 0.30s, 0.35s and 0.40s at DLBOW40°, the depth of the keyhole is not as deep as at 30° due to the collapse of the keyhole. It indicates that the keyhole is developed more easily as the laser incidence angle increases. For the DLBW process, the maximum keyhole depth is 4.02 mm, the minimum depth is 1.10 mm, and the average depth is 2.45 mm. This indicates that the actual heat input to the molten pool is higher in this process. The keyhole is unstable and highly susceptible to pore.



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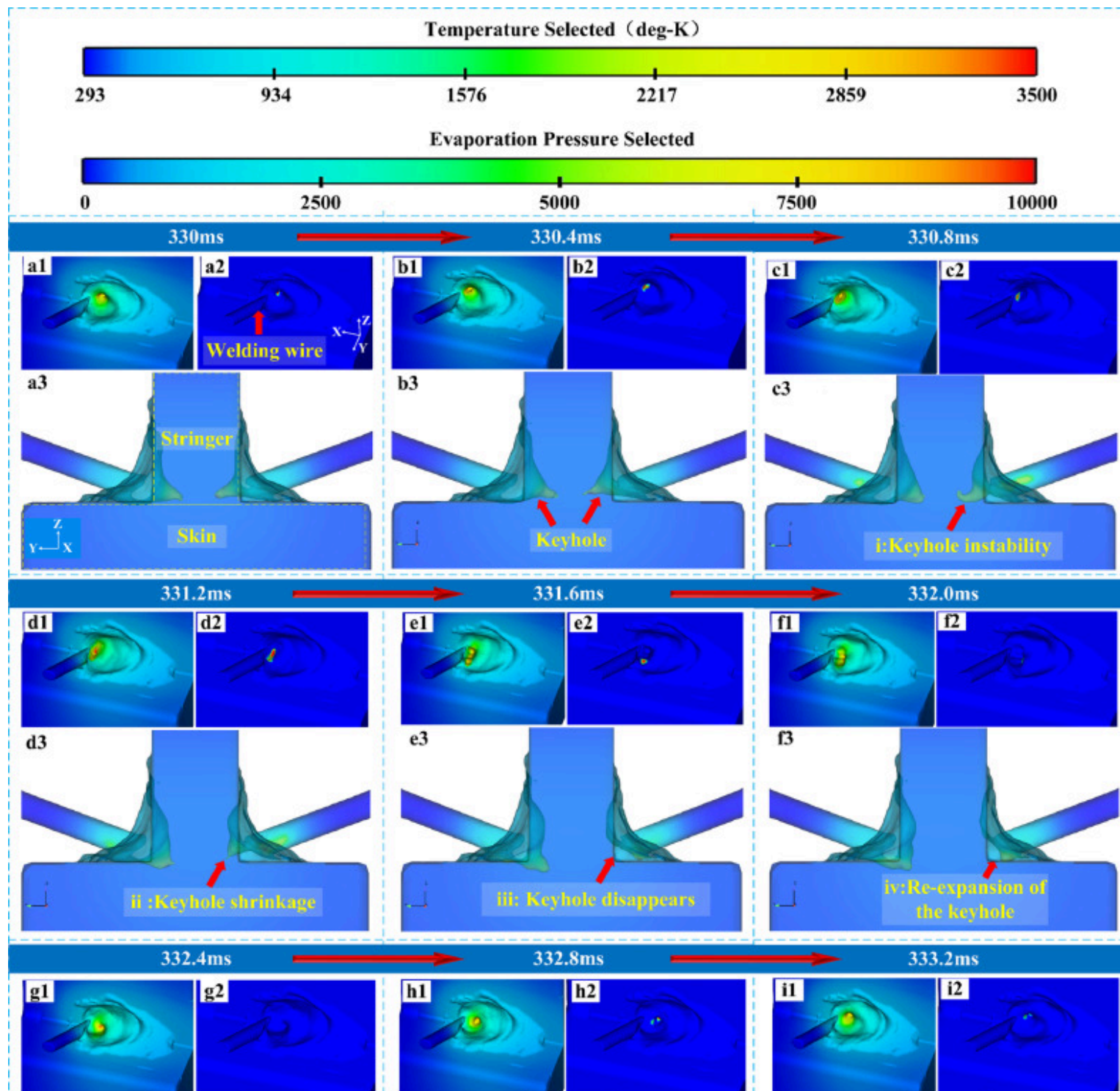
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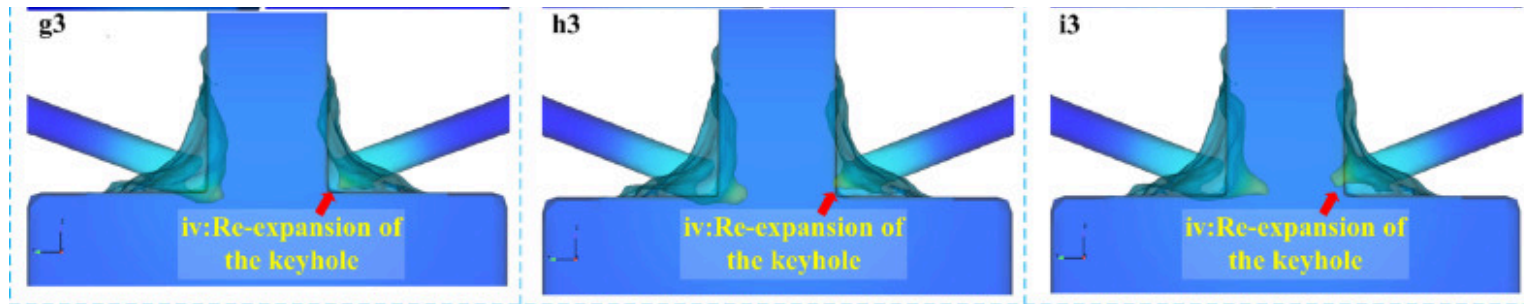
Fig. 8. Variation of keyhole depth at different moments under DLBOW20°, DLBOW30°, DLBOW40° and DLBW20°

4.3. Keyhole evolution under DLBOW process

4.3.1. DLBOW 20°

As in Fig.9 (a~c), the laser beam at an angle of 20° with the skin is moving counterclockwise from the stringer to the skin, at this time the keyhole is located on the stringer side. The keyhole is maintained under the direct action of the laser beam, where the temperature is higher and the recoil pressure is stronger. This is the period of maximum keyhole depth within one beam motion cycle. In Fig.9 (c3), instability was found at the bottom of the keyhole. In Fig.9 (d~f), the laser beam moves to the lowermost part of the circle, where the keyhole is located at the junction of the skin and the stringer. Because of the small tilt angle of the laser beam, the laser affects directly on the skin (Fig.9 (d1~f1)). Therefore, the beam energy is absorbed by the skin and the wire, resulting in a reduction of the recoil pressure used to maintain the keyhole shape, and the keyhole gradually shrinks (Fig.9 (d3)) to disappear (Fig.9 (e3)). It is not until the beam moves upward from the circular path that the keyhole expands again (Fig.9(f3)). In Fig.9 (g~i), the laser beam moves from the skin to the stringer and the laser energy is able to concentrate on the stringer. The keyhole is in the stage of re-expansion and evolves to the maximum depth. Then it enters the cycle of (Fig.9(a)) again. At an angle of laser incidence of 20° , the keyhole demonstrates a characteristic of large aperture and small depth. With the periodic evolution of the laser beam along its circumferential motion, the evolution rarely causes the formation of bubbles, which shows consistency with the experimental results.





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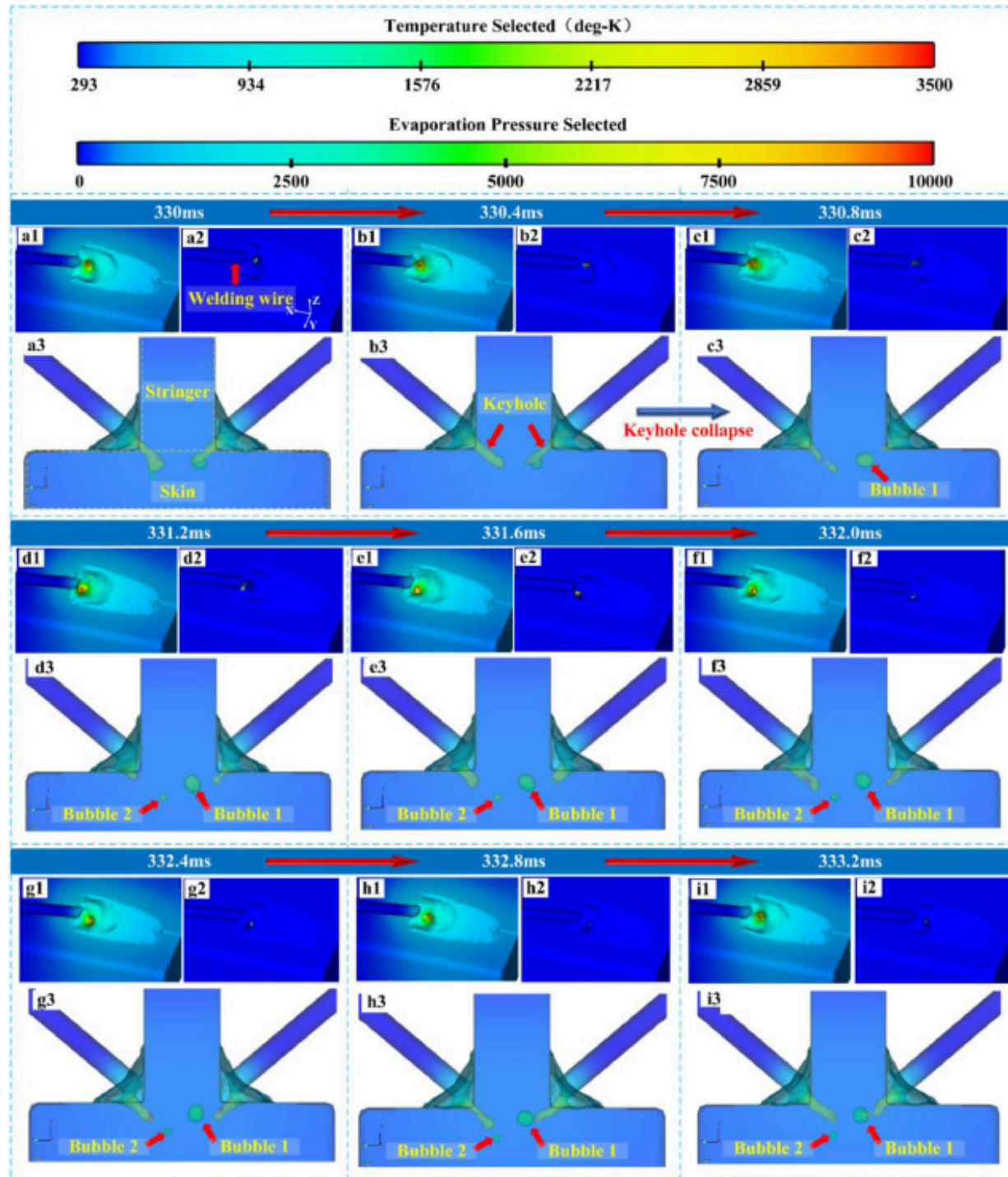
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Fig. 9. Evolution of the keyhole during one beam oscillation period at DLBOW 20°: (a1) ~ (i1) :Forming and temperature field of the side weld; (a2) ~ (i2) :Recoil pressure field for side welds; (a3) ~ (i3) :Evolution of the bilateral keyhole during a period.

4.3.2. DLBOW 40°

As shown in Fig. 10 (a~b), the laser beam with the skin angle of 40° moves counterclockwise from the stringer side to the skin side, and the keyhole is located on the stringer side at this time. The keyhole on both sides of the molten pool is more elongated and tilted in agreement with the angle of incidence of the laser beam. In Fig. 10(c), the laser beam moves to the junction of the skin and stringer, similar to 20°, the keyhole is destabilized at this position. Due to the slimmer keyhole at 40°, the right keyhole collapses to form bubble 1. In Fig. 10 (d~g), the keyhole is located on the skin side and the left keyhole in Fig. 10 (d) also collapsed to form bubble 2. Unlike the laser incident angle of 20°, where the laser beam acts directly on the skin and wire, resulting in a smaller actual heat input to the molten pool. Due to the larger laser incidence angle at 40°, the laser is able to focus directly on the molten pool and the keyhole expands rapidly towards the bubble. As shown in Fig. 10 (g3~i3). The laser beam moves

counterclockwise upward from the skin to the stringer side and the right keyhole is able to capture the bubble1 again. However, the left keyhole is located above and away from the bubble2, so the bubble2 on the left side is likely to be captured by the solidification front to form the pore. This phenomenon indicates that at an incidence angle of 40° , the bubbles are generated at the bottom of the molten pool and have a longer escape path, meaning that the possibility of blowholes is greater. At the laser incidence angle of 40° , the keyhole shows the characteristics of small caliber and large depth, evolving periodically with the circular motion of the laser beam. The evolution process is prone to bubbles, which may be captured by the solidification front to form a pore.



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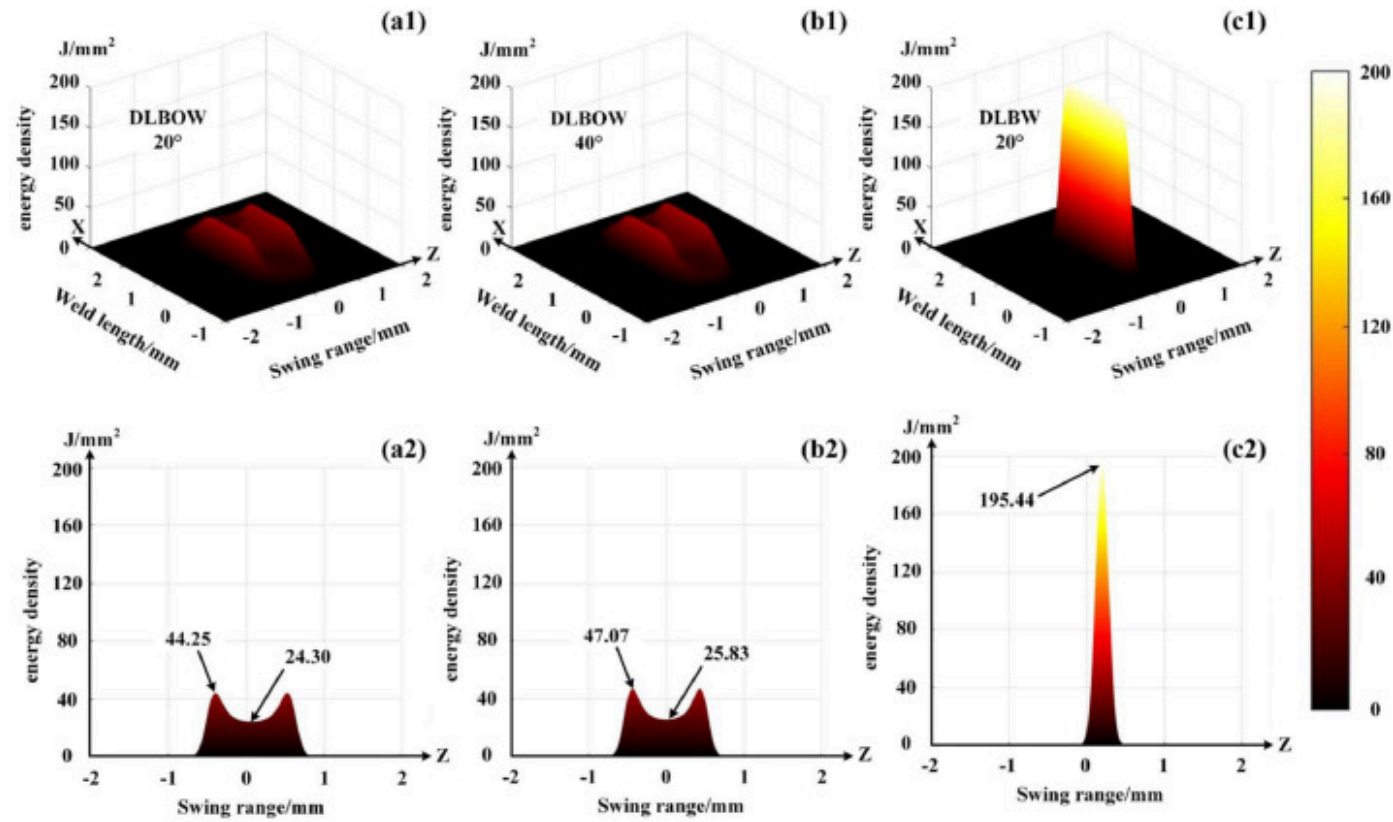
Fig. 10. Evolution of the keyhole during one beam oscillation period at DLBOW 40°: (a1) ~ (i1) :Forming and temperature field of the side weld; (a2) ~ (i2) :Recoil pressure field for side welds; (a3) ~ (i3) :Evolution of the bilateral keyhole during a period.

Comparing the evolution of the keyhole at different laser incidence angles, it is revealed that at small angles, when the laser moves to the junction of the skin truss, the laser directly affects the skin and the wire, resulting in less actual heat input to the molten pool, which reduces the intensity of the molten pool flow and the impact of the molten pool on the keyhole is weaker. The keyhole is more likely to remain stable compared to a large laser incidence angle. In addition, because of the difference in laser incidence angle, the force on the keyhole wall changes, which leads to the difference in keyhole morphology at different incidence angles. The mechanism of the force on the evolution of the keyhole at different laser incidence angles will be revealed in [Section4.5](#).

4.4. Effect of laser energy distribution and molten pool overheated zone on keyhole formation and molten pool flow

4.4.1. Energy density distribution of the laser

According to the melting behavior of the welding wire in [Fig.4](#) and the relationship between the laser beam and the T-joint in [Fig.5](#) and [Fig.6](#), it can be found that for the DLBOW20°, the plane of action of the laser beam lies in the XOZ plane where the vertical stringer is located, and the plane of action of the laser beam under DLBOW40° is perpendicular to the laser beam. The three-dimensional morphological characteristics of the beam energy density distribution at the corresponding action surface of the laser beam are calculated based on the heat source model and the laser motion path as in [Fig.11](#). The mathematical expression of the beam motion path is as follows:



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Fig. 11. Energy density distribution of the laser:(a1)~(a2):Axonometric and front view of DLBOW20°(b1)~(b2):Axonometric and front view of DLBOW40°(c1)~(c2):Axonometric and front view DLBOW20°.

DLBOW20°:

$$\begin{cases} x(t) = 2t - 0.05 \sin(2\pi ft) \\ z(t) = \frac{A}{2} \cos\theta \cos(2\pi ft) + d * \tan\theta + \frac{A}{2} \sin\theta \tan\theta * \cos(2\pi ft) \end{cases} \quad (22)$$

DLBOW40°:

$$\begin{cases} x(t) = 2t - 0.05 \sin(2\pi ft) \\ z(t) = -0.05 \cos(2\pi ft) \end{cases} \quad (23)$$

The laser energy density distribution characteristics under different paths (Eq(24)) are obtained by integrating Eq(2) in time over the beam motion paths Eqs ((22), (23)). Reference to two characteristic quantities[51]: One is average linear energy density E_{path} : E_{path} is the ratio of the total theoretical input energy to the total length of the path in a certain time, derived from Eqs ((24), (25), (26), (27)), which reflects the uniform effect of the great expansion of the scan trajectory on the energy distribution and the stirring effect on the molten pool. The second is the integrated peak energy density I_{max} , obtained by the Max function Eq(28), which indicates the maximum value of the laser energy density in the plane of action after taking account into the superposition effect of the spot, which reflects the degree of concentration of laser energy in the local area, and this parameter is related to the intensity of evaporation and boiling during the laser welding process.

$$E(x, y) = \int_0^T q(x, y, t) dt \quad (24)$$

$$E_{total} = \int_p^q \int_m^n E(x, y) dx dy \quad (25)$$

$$L = \int_0^t \sqrt{x^2(t) + y^2(t)} \quad (26)$$

$$E_{path} = \frac{E_{total}}{L} = \frac{E_{total}}{\int_0^t \sqrt{x^2(t) + y^2(t)}} \quad (27)$$

$$I_{max} = \text{Max}(E(x, y)) \quad (28)$$

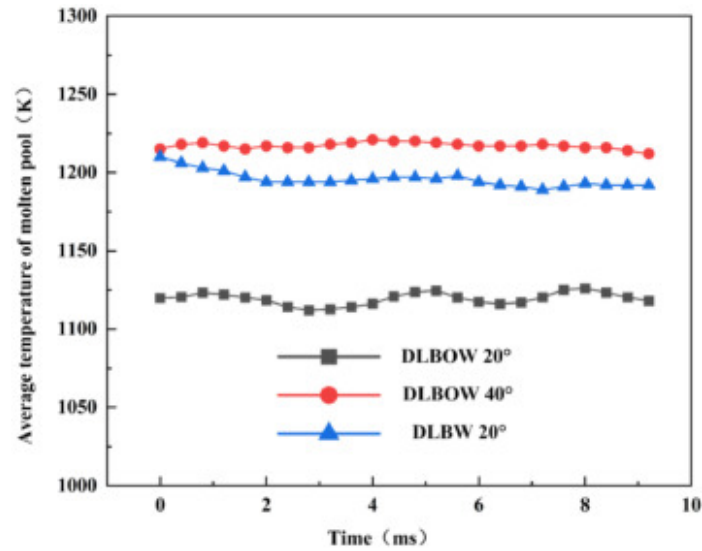
According to the results in Table 5, the I_{max} and E_{path} of 40° at DLBOW are 6.3% and 3.7% higher than those of 20°, respectively, and the laser energy distribution is more equal at DLBOW20°. The I_{max} and E_{path} of DLBOW are much lower than those of DLBW, indicating that the expansion of the trajectory length by the oscillation process dramatically reduces the laser peak energy density and the trajectory transient line energy, effectively homogenizing the laser energy density distribution and reducing the intensity of evaporation and boiling during the laser welding process.

Table 5. Characteristic parameters of energy distribution under different oscillation parameters

Oscillation mode	$E_{\text{total}}(\text{J})$	$I_{\text{max}}(\text{J}/\text{mm}^2)$	$E_{\text{path}}(\text{J}/\text{mm})$	L(mm)
DLBOW20°	210	44.25	2.15	94.35
DLBOW40°	210	47.07	2.23	97.40
DLBW20°	210	195.44	105	2

4.4.2. Molten pool temperature and overheated zone

Fig. 12 shows the average temperature variation curve of the molten pool at different moments. The molten pool temperature fluctuates little with time during the three beam oscillation cycles. Interestingly, the curve fluctuation period of DLBOW matches well with the beam oscillation period, especially 20° (around 3.33 ms), which may be related to the beam movement to different positions. The fluctuations of DLBW, on the other hand, do not have much regularity. Further, the temperature at each time is averaged to obtain the average temperature of the molten pool during this period, DLBOW20°:1119K; DLBOW40°:1217K; DLBW20°:1119K. Among them, the DLBOW 20° molten pool has the lowest average temperature, indicating that the actual heat input to the molten pool is lower at 20 oscillation. This is related to the laser energy density distribution, but there should be a more profound reason.



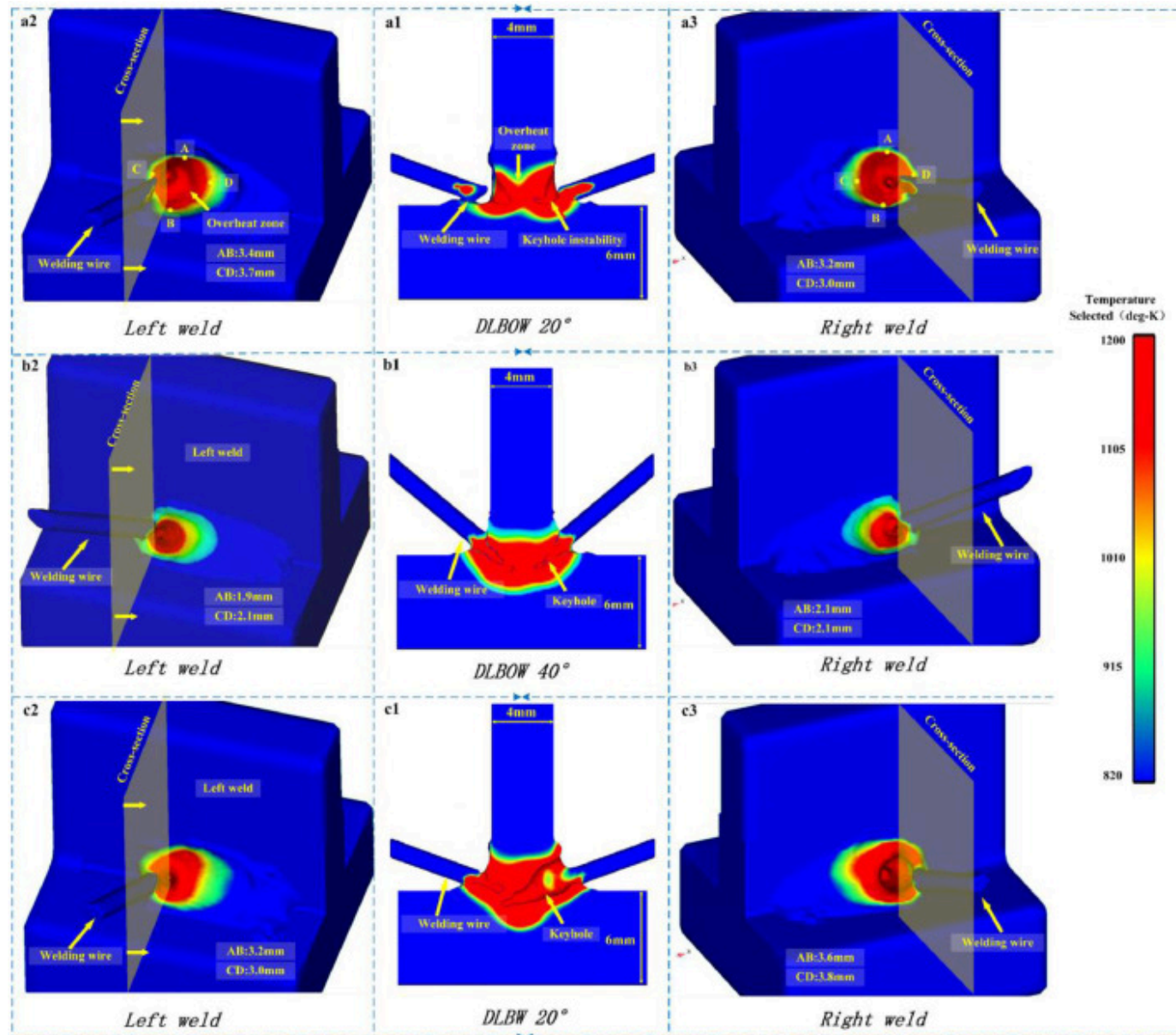
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Fig. 12. The average temperature of the molten pool at different moments under different oscillation parameters

As shown in Fig. 13, a cross section is taken at the keyhole of the weld. Red represents the area where the molten pool temperature exceeds 1200 K, while blue is the unmelted area below the melting point of 820 K. Compared to DLBOW40°, the superheat zone on both sides of DLBOW20° is larger but the cross-sectional is smaller. As in Fig. 13(a2, a3), take the two lines segments AB and CD, which are farthest apart from each other in the overheated zone, and use the average value of their multiplier S_{side} to define the size of the overheated zone of the lateral weld. Calculate the area of the overheated zone of the cross-section in Fig. 13(a1) $S_{positive}$.

$$S_{absorption} = \frac{S_{side}}{S_{positive}} \times 100\% \quad (29)$$



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Fig. 13. Overheat zone of the molten pool:(a1)~(a3),(b1)~(b3),(c1)~(c3):Overheat zone of the molten pool at DLBOW 20°,DLBOW 40°and DLBW 20°in the cross section and both sides of the weld

$S_{absorption}$ is the ratio of the area of the lateral overheated zone to the positive overheated zone. The larger the value, the more obvious is the characteristic of a large caliber and small depth of the overheated zone of the molten pool. The three value of DLBOW20°, DLBOW40° and DLBW20° are 76%, 26% and 50% respectively. As the increase of keyhole depth enhances the multiple laser reflections and heat transfer inside the keyhole, resulting in an increase of the actual heat input to the molten pool. Therefore, for DLBOW40°, the laser beam is absorbed by multiple reflections in the slender keyhole, the keyhole effect is more obvious, and the actual laser energy absorbed by the molten pool is more. Less laser energy is reflected to the opening, resulting in a smaller overheating area of the molten pool at the opening. In addition, due to the smaller overheated zone on both sides of the molten pool, the smaller possibility of bubbles escaping after floating, resulting in a greater tendency to form pore defects

4.5. Mechanism of keyhole evolution at different laser incidence angles

Fig. 14(a) shows the projection of the beam trajectory in the YOZ plane, with the axes of the laser beams on both sides at an angle of θ to the horizontal plane. The laser beams move around their respective axes in a circular motion. From Fig. 14(b), it can be found that with the increase of the angle θ , v_x remains constant, v_y increases and v_z decreases. This is due to the fact that the time for the beam to move from the low point A to the high point B in Fig. 14(a) remains constant. As the angle θ increases, the path of the beam moving in the z-direction becomes shorter. Therefore, v_z decreases, while v_x remains constant, then v_y increases. This means that when the laser beam is incident at an angle of 20°, v_z is bigger, the impact of gravity in the Z direction is stronger, and the keyhole is more difficult to maintain.



Fig. 14. Mechanism of keyhole evolution at different laser incidence angles:(a):Laser movement trajectory in YOZ view;
(b):Velocity components in Y and Z directions with different oscillation parameters;(c):Keyhole evolution;(d):The characteristics of keyhole variation under different oscillation parameters.

In Fig. 14(c) The quasi-static analytical model[52] is used to establish the force relationship for the local location of the upper keyhole. Where, r is the radius of curvature, reflecting the contraction of the keyhole in the local area, and τ is the vapor-metal friction force. Recoil pressure P_R supports the keyhole wall outward, while surface tension P_s and the gravitational component $G\cos\theta$ perpendicular to the wall of the keyhole are directed inward, driving the keyhole to collapse and close. These four forces collectively contribute to maintain the quasi-static state of the keyhole.

For the oscillating process, when the laser incidence angle θ is small, the gravitational component $G\cos\theta$ perpendicular to the wall of the keyhole increases due to the decreases of θ , and at the same time, a portion of the energy of the laser beam is dispersed to function on the skin and the wire, thus the recoil pressure P_R is lower. As a result, it is difficult for the keyhole to expand further into the molten pool along the direction of laser beam incidence, hindered by surface tension and gravity. At the upper wall of the keyhole with an inclination angle of θ , only a small surface tension P_s is needed to maintain the quasi-static of the keyhole because the molten pool temperature is lower, the recoil pressure P_R is smaller, and $G\cos\theta$ is larger at this time. Since P_s is inversely proportional to the keyhole radius r , the r of the keyhole wall is larger. As a result, DLBOW20° has small keyhole depth and large caliber. In summary, the stable keyhole evolution process leads to less possibility of bubbles generation, secondly, the average depth of keyhole is shallow and the bubbles generated has a shorter uplift distance, finally, the DLBOW20° molten pool is relatively smooth and has a large superheated zone on both sides of the molten pool, which makes it easier for bubbles to escape. Therefore, the porosity of DLBOW20 is lower.

When the laser incidence angle θ is larger, $G\cos\theta$ decreases due to the increase of θ . Meanwhile, according to the previous analysis, it is known that the laser beam energy density is higher and more concentrated because the laser effects directly on the molten pool. Therefore, the keyhole is more easily extended into the molten pool along the laser incidence direction. With the more pronounced keyhole effect, multiple laser reflections and increased heat transfer within the keyhole lead to higher temperatures in the molten pool and keyhole, and therefore higher recoil pressure P_R . As in Fig. 14(c), in order to maintain quasi-static, the keyhole radius r is reduced to meet the larger surface tension P_s . In summary, for DLBOW 40°, firstly, the slender keyhole is more prone to destabilization by the impact of the molten pool and the contraction of the larger surface tension to form bubbles, Secondly, the bubbles generated have a longer uplifting path because the keyhole is relatively slender, Finally, it is more difficult for bubbles to escape because the molten pool flow is more turbulent and the overheated zone on both sides of the pool is smaller. Therefore, DLBOW 40° has a higher porosity.

For processes without oscillation, due to the high laser energy density, laser action on the molten pool causes rapid temperature rise, vaporization, and sublimation of the molten pool. The recoil pressure P_R at the bottom of the keyhole is high. However, since the laser beam is radiated only in part of the keyhole under the single laser process, the side walls of the keyhole tend to collapse under the effect of $G\cos\theta$ and surface tension P_s to form bubbles. Due to the large tilt of the keyhole at a larger laser incidence angle θ , the bubbles generated at the bottom of the molten pool escape farther and have a greater tendency to be captured by the solidification front to form a porosity.

5. Conclusions

In this work, a new process of dual-laser beam oscillating bilateral synchronous welding (DLBOW) was proposed for the problem of pores easily generated in laser welding of T-joints, which successfully suppressed the pore defects in the weld seam. The corresponding CFD numerical model was established, and the simulation results showed good agreement with the experimental results. The molten pool flow characteristics and keyhole evolution of DLBOW and DLBW were compared and analyzed to reveal the mechanism of pore formation by different processes and different laser incidence angles. Found that for T-joint dual-laser beam bilateral synchronous welding (DLBOW), the oscillation process can homogenize the laser energy, reduce the peak molten pool temperature, slow down the molten pool fluctuation, and significantly reduce the porosity. In addition, the molten pool is symmetrical on both sides of the stringer for different welding parameters and the tilting angle of the keyhole is consistent with the laser incidence angle. The weld depth is shallower and the molten pool is smoother for smaller incidence angles. Finally, In the oscillation process, the keyhole is susceptible to instability when the laser moves to the junction of the skin and the stringer. smaller laser incidence angles result in shorter escape paths, lower tendency for bubble generation and smaller porosity of the weld.

Author contributions

Y.Y.H. and P.Y.X. conceived the idea and designed the experiments. P.Y.X., X.S.F., G.Y.M and C.M.W conducted the welding experiments and fabricated the 2219 joint. Y.Y.H, X.Z., C.C.Z and L.D.X developed a numerical model and performed post-processing analysis. Y.Y.H and Z.C.C Processed data and wrote articles. Y.Y.H. and M.Y.Z supervised the whole work. All authors contributed to reviewing and editing the manuscript and discussing and interpreting all the results.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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